Student Eats — Solution Architecture

Three-Tier Web Application for Student Meal Ordering

PRESENTATION TIER

APPLICATION TIER

DATA TIER

Student User
Mobile / Desktop
Browse • Order • Pay

Seller
Vendor Portal
Manage • Fulfill • Notify

Administrator
Admin Dashboard
Monitor • Moderate

HTTP/HTTPS Requests (JSON)

WEB FRONTEND (React / Vue)

- **Auth**
  Login / Register
  JWT Token
  OAuth
- **Marketplace**
  Browse Meals
  View Details
  Reviews
- **Search & Filter**
  By Category
  By Price
  By Rating
- **Cart**
  Add Items
  Modify Qty
  Remove
- **Checkout**
  Payment
  Delivery Info
  Confirm

SELLER DASHBOARD

- **Meal Management**
  Add / Edit / Delete Meals
- **Order Management**
  View & Update Status

REST API Calls

BACKEND API (FastAPI / Python)

- **FastAPI**
  REST Server
  Pydantic Models
  Async I/O
  OpenAPI Docs
- **Auth Service**
  JWT Handler
  Password Hash
  Role-based
  Session Mgmt
- **Meal Service**
  CRUD Meals
  Image Upload
  Categorization
  Availability
- **Order Service**
  Create Order
  Status Tracking
  Payment Link
  History
- **Notification**
  Email Alerts
  Push Notifs
  SMS Gateway
  WebSocket
- **Review Service**
  Post Review
  Rating Calc
  Moderation
  Analytics

SQL Queries via ORM (SQLAlchemy)

SQLITE DATABASE

- **Users** — Students
- **Sellers** — Vendors
- **Meals** — Menu Items
- **Orders** — Transactions
- **Reviews** — Ratings

External APIs
Payment Gateway
Email Service
SMS Provider

Legend

- Request Flow
- Response Flow
- Client / Actor
- Frontend Module
- Backend Service
- Database Table

Technology Stack

Frontend
React / Vue.js / Tailwind CSS
Axios (HTTP Client)

Backend
FastAPI (Python 3.11+)
SQLAlchemy ORM + Alembic
Pydantic Validation
Uvicorn ASGI Server

Database
SQLite (Development)
PostgreSQL (Production)

Infrastructure
Docker / Nginx / HTTPS

Student Eats — Capstone Project Architecture Diagram